

Fig. 65.1 Rationale and role of therapeutic plasma exchange for reducing end-organ injury. Pathologic conditions result in accumulation of toxic proteins, antibodies, or other molecules that induce tissue damage directly and through the associated inflammatory response. A two-pronged approach to slow or arrest this process is to 1. remove toxic molecules when feasible (top green box) and 2. suppress production of these substances with appropriate immunosuppressive medication (lower green box). Corticosteroids are both immunosuppressive and anti-inflammatory.

Fig_65_1

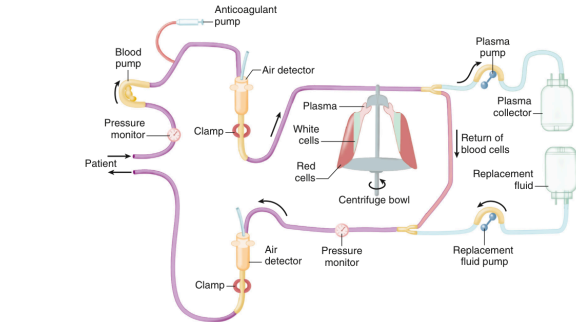


Fig. 65.2 Centrifugal separator for plasma exchange. Blood is pumped into the separator container. As the centrifuge revolves, different blood components are separated into discrete layers, which can be harvested separately. Plasma is pumped out of the centrifuge into a collection chamber. Red blood cells, leukocytes, and platelets are returned to the donor, along with replacement fluid. (From Madore F, Lazarus JM, Brady HR. Therapeutic plasma exchange in kidney diseases. *J Am Soc Nephrol.* 1996;7:367-366.)

Fig_65_2

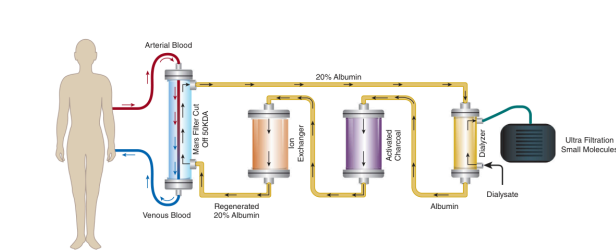


Fig. 65.3 Molecular Adsorption Recirculating System (MARS). MARS provides enhanced toxin removal by combining traditional removal of water-soluble toxins through conventional hemodialysis and albumin-bound toxins by linking an albumin circuit to remove protein-bound toxins. Blood is pumped from the patient into the MARS flux dialyzer that has a 10% to 20% albumin dialysate flowing countercurrent at ~600 mL/min. The MARS dialyzer is porous for molecules up to 50kDa, so it restricts growth factors and albumin from freely crossing the membrane, but smaller protein-bound toxins filter through and the albumin dialysate has capacity to bind the filtered toxins. The albumin dialysate is then sequentially circulated through a charcoal filter and ion exchange resin before the "cleaned" albumin dialysate is returned to the patient. The standard continuous renal replacement therapy filter is used for ultrafiltration and removal of water-soluble toxins.

Fig_65_3

Table 65.1 Pathologic Factors Removed by Therapeutic Plasma Exchange

Autoantibodies
Immune complexes
Myeloma proteins
Cryoglobulin
Complement products
ADAMTS-13 (metalloproteinase)
Lipoproteins
Protein-bound toxins

Table_65_1

Table 65.2 Summary of Kidney Diseases Treated With Plasmapheresis

Disease	Category
Antiglomerular basement membrane disease	I
Rapidly progressive glomerulonephritis	II
Hemolytic uremic syndrome	III
TTP	I
Renal transplant rejection	IV
Desensitization for renal transplantation	II
Recurrent FSGS	III
Cryoglobulinemia	II
Systemic lupus erythematosus	III

Category I, Standard primary therapy; Category II, supportive therapy; Category III, when the evidence of benefit is unclear; Category IV, when there is no current evidence of benefit or for research protocols; FSGS, focal and segmental glomerulosclerosis; TTP, thrombotic thrombocytopenic purpura.

Table_65_2

Table 65.3 Renal Recovery According to Initial Creatinine Concentration in Treated Patients With Anti-Glomerular Basement Membrane Antibody Disease

Study, Year (Reference)	Patients (No.)	Patients With Independent Renal Function at 1 Year	Treatment
		Initial Cr Concentration <600 mol/L (<6.8 mg/dL)	Initial Cr Concentration ≥600 mol/L (≥6.8 mg/dL) or dialysis dependent
Bouget et al., 1990	13	50%	0%
Herody et al., 1993	29	93%	0%
Merkel et al., 1994	32	64%	3%
Andrews et al., 1995	15	NA	7%
Daly et al., 1996	40	20%	0%
Levy et al., 2001	71	94%	15%
Saurina et al., 2003	32	71%	18%
Cui et al., 2011	176	25%	3%
Huat et al., 2016	122	40%	2.4%
Zahir et al., 2021	48	4%	8%

C, Corticosteroids; CFM, cyclophosphamide; Cr, Creatinine; PE, plasmapheresis.

Table_65_3

Table 65.4 Indications for Plasma Exchange in Pregnancy-Related Conditions

Condition	Indication	Additional Notes
Thrombotic thrombocytopenic purpura (TTP)	Removes ADAMTS13 autoantibodies and replenishes ADAMTS13 enzyme with fresh frozen plasma (FFP)	A life-threatening emergency requiring prompt treatment. Typically combined with corticosteroids
Hemolysis, elevated liver enzymes, low platelets (HELLP)	Severe cases with multorgan failure or atypical presentation	Limited evidence; TPE may be beneficial in refractory cases
Severe hypertriglyceridemia in pregnancy	When severe hypertriglyceridemia leads to acute pancreatitis	TPE removes triglycerides and prevents further organ damage
Autoimmune diseases (e.g., lupus, myasthenia gravis)	Treats severe flares, especially in life-threatening situations like lupus nephritis or myasthenic crisis.	TPE helps in antibody removal; adjunctive to immunosuppressive therapy
Refractory preeclampsia or eclampsia	Limited use in severe, refractory cases with multorgan dysfunction	Insufficient evidence; only in exceptional situations and after delivery
Severe Rh isoimmunization	Reduces circulating maternal alloantibodies targeting fetal red cells	Typically reserved for severe cases where fetal transfusions are not sufficient

Table_65_4